

Child-Involved Group Prioritization: Effects of Group Composition on Experience, Participation Level, and Interaction

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Abstract

Stakeholder-informed prioritization (SIP) is a lightweight technique designed to help child-involved groups quickly understand and prioritize research and design directions. However, outcomes of short prioritization discussions can vary with group composition. In this paper, we deployed SIP with 34 groups of children and parents ($N = 94$) at a large community event, focusing on how group composition, including group type (family-based vs. child-based) and familiarity (mixed-familiarity vs. fully familiar), shapes SIP experiences. Results suggest that child-based groups were associated with higher satisfaction, more active child participation, smaller in-group range in comfort, satisfaction, and participation level, and more verbal reasoning. Fully familiar groups reported higher comfort and showed more verbal reasoning and less facilitator involvement, while mixed-familiarity groups showed a smaller in-group range in participation levels. We discuss implications for designing and facilitating child-involved group prioritization sessions and provide practical guidance for using SIP as a rapid method in early-stage research.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Children, Child-Involved Group, Group-based Research, Prioritization, Focus Groups

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1 Introduction

Today's technologies are becoming a bigger part of children's daily lives, making it important to include their perspectives in research and design to reflect their needs, perspectives, and preferences [17, 35]. Participatory Design (PD) is a common approach in the Interaction Design and Children (IDC) community that involves children as end-users in the design and research process [34]. Unlike approaches that treat end-users as passive informants, PD emphasizes co-design, where children contribute ideas, provide feedback, and help shape design outcomes across the process [10]. However, in many cases, children's influence on research priorities can still be limited, with participation often centered on reacting to pre-defined concepts rather than shaping the direction itself [18, 36].

Researchers also use stakeholder-informed prioritization methods such as focus groups, interviews, and card sorting to surface needs and make early decisions. When applied to children, these methods can be difficult to run well. They often rely on structured deliberation, ranking, or consensus-building that may not match children's communication styles or developmental stages [27, 33]. Intergenerational settings can introduce power differences that unintentionally sideline children's voices, leading to outcomes that do not fully reflect their perspectives [9, 36]. Children's attention span, their ability to discuss abstract tradeoffs, and differences in confidence can further shape who speaks, how much, and whose ideas carry weight [14, 41]. Facilitators can reduce these barriers, but facilitation alone cannot fully address a core issue: even with the same activity, group outcomes may depend strongly on who is in each group.

This paper focuses on the effects of group composition in short child-involved prioritization discussions. We use Stakeholder-Informed Prioritization (SIP) [22], a lightweight, structured group activity for discussion and ranking, as a standardized study setting. SIP is a technique designed to support rapid prioritization in early-stage work,

including time-limited contexts (e.g., community events) where researchers want both an explicit prioritized outcome and the reasoning that emerges through discussion. Because every group completed the same structured SIP activities, we could compare groups within a shared activity context and examine whether differences in experience, participation, and interaction patterns were related to group composition.

We analyze a de-identified dataset from a prior SIP deployment in a Virtual Reality (VR) case context [22]. The original work introduced SIP and reported overall feasibility and outcomes, while this paper focuses on how group composition shapes SIP experiences, participation, and interaction patterns. Specifically, we used data that included SIP sessions from 34 groups of children and parents ($N = 94$), collected at a large community event. We examine two composition factors: **group type** (family-based vs. child-based) and **familiarity** (mixed-familiarity vs. fully familiar).¹ We ask the following research questions:

- **RQ1 (Experience):** How do group type and familiarity relate to perceived experience in SIP sessions (comfort, inspiration, and satisfaction)?
- **RQ2 (Participation):** How do group type and familiarity relate to participation level within groups?
- **RQ3 (In-Group Range):** How do group type and familiarity relate to in-group range in perceived experience and participation?²
- **RQ4 (Forms of interaction):** How do group type and familiarity shape interaction patterns during SIP discussions (e.g., how groups justify choices, negotiate disagreements, and sustain discussion)?

Across measures, we find that group composition matters. Child-based groups were associated with higher satisfaction, more active child participation, smaller in-group ranges in comfort, satisfaction, and participation, and more verbal reasoning. Familiarity showed mixed patterns: fully familiar groups reported higher comfort, showed more verbal reasoning, and required less facilitator involvement, while mixed-familiarity groups showed a smaller in-group range in participation.

This paper contributes to the IDC communities in two ways:

- We provide empirical evidence on how group composition (group type and familiarity) shapes experience, participation, in-group range of perceived experiences and participation, and interaction patterns in child-involved prioritization discussions.
- We offer practical guidance for recruiting and facilitating child-involved group prioritization sessions using SIP as a structured activity, with attention to supporting youth voice in both child-only and intergenerational groups.

¹Fully familiar means all group members knew each other, while mixed-familiarity means a mix of people who knew and did not know each other.

²This examines whether some members had a substantially worse experience or contributed substantially less than others.

2 Related Work

2.1 Group Composition in Child-Computer Interaction

Current research on ideal group composition in research with children is mixed. Best practices often emphasize the balance between homogeneity and heterogeneity within groups. Homogeneous groups tend to promote a sense of psychological safety and reduce social inhibition [32, 42], while the downside is that it can limit the diversity of perspectives, potentially resulting in more uniform or less critical discussions [10, 32]. For example, Porcellato et al. [32] conducted 12 focus groups and concluded that smaller, homogeneous groups, such as single-sex or same-age children, are recommended to decrease discomfort and distractions [42]. However, one limitation is that discussions in these groups can rely more on facilitator-child interactions, limiting peer-to-peer exchange. Another reflection [3], based on field notes and qualitative focus group data, showed how recruiting strangers versus families affected group dynamics, emphasizing the advantages of holding focus groups in natural settings where familiar participants engage more openly.

Common group setups involve (1) separating families to engage with parents and children separately [11, 21, 28] and (2) family-based groups of parents and children together [5, 6]. Each of these group setups comes with its own tradeoffs. Previous work has found that designing solely with parents can be useful to think about specific interactions such as parental controls [29], but children largely have very dissimilar needs and values to adults [11]. Meanwhile, participating in children-only groups raises concerns about comfort and children feeling “observed” by an adult researcher [34]. Family-based groups offer additional insights, as they allow for the exploration of family dynamics and collective behaviors [5], such as physical activity patterns [4]. Therefore, similar to heterogeneous groups, family-based groups are associated with participation imbalances, where the adult voices overshadow the youth voices [19]. Child-computer interaction methods must strike an appropriate balance between consulting parents and engaging with children.

Inspired by these mixed findings and tradeoffs, we examine how two concrete composition factors, *group type* (child-based vs. family-based) and *familiarity* (mixed-familiarity vs. fully familiar), shape short child-involved prioritization discussions, focusing on experience, participation balance, and interaction patterns.

2.2 Stakeholder-Informed Prioritization (SIP)

Children are important when forming design and research directions for them and their families, as they are directly impacted by the outcomes and often actively engage with them. A significant body of research highlights the value of children as informants, where they contribute as information providers to guide research and technology design directions, as defined within Druin’s participatory design (PD) framework [12, 39]. Stakeholder-Informed Prioritization (SIP) is one lightweight technique that supports short discussions where participants compare alternatives and produce a ranked outcome [22]. SIP was introduced to help researchers gather early-stage priorities from multiple small groups within limited

time windows (e.g., public or community events), while still capturing the reasoning that emerges during group discussion. Compared to longer participatory approaches that rely on iterative co-design cycles [10, 34], SIP is designed for rapid elicitation: groups can quickly understand a set of candidate directions, discuss tradeoffs, and externalize decisions through a shared ranking artifact.

A core feature of SIP is its use of a fixed task structure and shared materials (e.g., alternative tokens arranged on a ranking surface) that make the prioritization process visible and discussable. This artifact-mediated format can reduce reliance on abstract talk alone and can support children's participation by providing concrete referents for negotiation and explanation. Because SIP maintains a consistent task format across groups, it can also serve as a standardized setting for comparing how group-level factors shape discussion processes. Building on prior work that introduced SIP and demonstrated its feasibility as a rapid prioritization activity [22], the present study uses SIP as a common task to examine how group composition relates to experience, participation balance, and interaction patterns in child-involved prioritization discussions.

3 Methods

To answer our research questions and provide actionable insights to guide SIPs in IDC research, we analyze SIP sessions from 34 groups ($N = 94$) collected at a large community event. In this section, we describe the SIP materials and study design that support comparisons of experience, participation, and interaction across different group compositions.

3.1 Identifying Alternatives and Creating Materials

This study used a Virtual Reality (VR)-in-the-home case context to ground the prioritization task. The alternatives were derived from a secondary analysis [24] of data collected in our prior empirical study on children's and parents' perspectives on VR in the home [23]. Following the SIP protocol [22], researchers prepared eighteen pre-selected alternatives. Because this number is too large for a single short session, researchers organized them into two SIP boards (details can be found in the Appendix A): nine empirical directions and nine engineering/design directions. Each alternative was represented as a custom-printed magnet, and researchers also included dry-erase magnets to allow participants to add new alternatives. During the session, groups placed magnets on a physical whiteboard to create a ranked list.

3.2 Recruitment Strategy, Settings, and Participants

Researchers deployed the study for three days at the Driven to Discover research facility at the Minnesota State Fair, a public community event. A dedicated booth at the event served as a "lab-in-the-wild" designed to engage diverse attendees (Fig. 1). Recruitment took place as families passed the facility, and researchers invited children eight and older and their grown-ups to share their opinions about VR in the home.

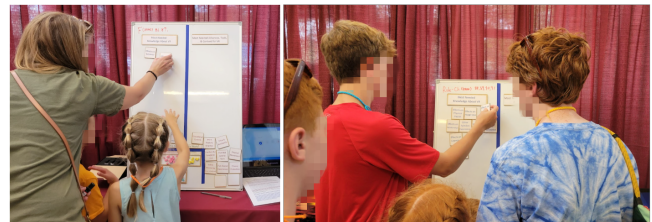


Figure 1: Example SIP sessions by group type: Family-based group (left); Child-based group (right).

This study used data from 94 participants (29 parents, 65 youths) from 34 groups.³ All demographic questions were optional. Among the parents who reported age ($N = 29$), the average age was 47.0 ($SD = 10.3$). For parents who reported gender ($N = 28$), there were 14 females and 14 males. Among those reporting ethnicity ($N = 29$), 22 identified as White, 4 as Latinx, and 3 as other ethnicity. Among the adults who reported socioeconomic status (SES) factors ($N = 25$), 2 did not hold a college degree, and 1 reported annual income below \$75,000. Youth participants had an average age of 13.4 ($SD = 3.1$), with 30 females, 29 males, and 2 identifying as another gender. Of the children who reported ethnicity ($N = 61$), 44 identified as White, 1 as Black, 5 as Latinx, and 11 as other.

The study featured two conditions to explore SIPs and forms of interactions with different group compositions. Participants were recruited into groups of 2-4 members. In one condition (family-based), each family (at least one parent and one child) participated as a group. In the second condition (child-based), children were grouped by role working together. Assignment to condition was based largely on participant availability and interest. For example, if only one parent and their children were ready to do the study, they would participate in the study as a family group (all members were in the same family group). If there was a group of youths who wanted to do the study but their adult did not, they would be grouped into a child-based group. Data from 34 groups were used in this analysis: 20 family-based groups ($N = 58$) and 14 child-based groups ($N = 36$). Within the family-based groups, there were 29 parents and 29 children; all group members knew each other. In child-based groups, participants did not necessarily know each other beforehand. Some child-based groups consisted of all people who knew each other (full-familiarity) (9 groups, $N = 21$), while others had a mix of people who knew and did not know each other (mixed-familiarity) (5 groups, $N = 15$). We did not further examine familiarity within mixed-familiarity groups, as our primary focus was on naturally occurring group compositions rather than artificially creating mixed-familiarity conditions. The average family-based group size was 2.9 ($SD = 0.81$), and the average child-based group size was 2.57 ($SD = 0.85$). A Mann-Whitney U test found no significant difference in group size between two group types.

³The original deployment recruited 124 participants [24]. For the present analysis, we excluded groups whose audio recordings were unusable or incomplete because of technical issues or background noise.

3.3 Data Collection Procedure

Researchers obtained written informed consent from participants 14 and older, written parental consent from parents for children under the age of 14, and verbal assent from children under 14 (as reviewed and approved by our IRB). Participants shared their demographic information and watched short scenario videos about the potential use of VR at home (scenarios came from a prior VR ethics study for children [23], see Appendix B). Participants were then grouped as described in the prior section and invited to work together to discuss and rank alternatives. Researchers audio-recorded the sessions, took notes, and helped manage time. This group activity lasted about 15 minutes ($M = 9.2$ mins, $SD = 3.4$ mins). The audio recordings were transcribed into text for analysis.

After the SIPs, researchers distributed a paper questionnaire, which included three statements using a 5-point Likert-type scale from “strongly disagree” to “strongly agree”:

- Q1 (Comfort): I felt comfortable sharing my thoughts with my group.
- Q2 (Inspiration): My group helped me consider issues in a new way.
- Q3 (Satisfaction): I am satisfied with the final results.

After completing the study, participants were offered a backpack or toy as compensation. A total of four researchers were trained as facilitators to explain the task using a standardized script. Beyond the task introduction, researchers intervened only if participants explicitly requested assistance or if the entire team was unable to progress and became stuck.

3.4 Data Analysis

We obtained measures for the quantitative analysis in this study to answer our research questions, as shown in Table 1.

3.4.1 Perceived Comfort, Inspiration and Satisfaction (RQ1). In order to understand how different group compositions (e.g., group types, familiarity) influenced participant experiences, we analyzed individual responses to the three Likert-scale questions. Since individual responses are not independent (i.e., influenced by the specific group and by conditions) and violate the normality assumption (Anderson-Darling Test), we used a Mixed-Effects Ordinal Logistic Model for analysis. This model was chosen for several reasons: first, it accounts for the nested structure of the data, where individual responses are grouped within families or roles; second, it allows for the inclusion of both fixed effects (e.g., group type) and random effects (e.g., variations within each group); and third, it is robust to violations of normality in the residuals, making it appropriate for analyzing non-normal data.

3.4.2 Level of Participation (RQ2). To measure participation level, we calculated individual word counts as an indicator of verbal contribution. We fitted a linear mixed-effects model with group type and familiarity as fixed effects and group ID as a random intercept to account for the nesting of participants within groups. This model allowed us to estimate differences in the number of words spoken across group compositions.

3.4.3 In-Group Range of Perceived Experiences and Participation (RQ3). To evaluate whether some members of a group had substantially worse experiences or contributed substantially less than others, we analyzed aggregated group-level measures. For perceived experiences, we calculated the within-group range of questionnaire responses for Q1–Q3. For participation, we calculated each participant’s word count as a proportion of the total group word count, and then calculated the within-group range of these proportions. Because these range measures can include zero values, we used GLMs with a Gaussian distribution and identity link.

3.4.4 Forms of Interaction (RQ4). We used a content analysis approach [13] to quantitatively analyze the forms of interactions under different group compositions. The first author developed an initial codebook based on inductive insights from the data by closely reviewing a subset of the transcripts (11.76% of the data), identifying recurring patterns and themes without preconceived categories. These insights were refined through group discussions with the research team and combined into a single codebook. Each code in the codebook was defined, with descriptions and examples provided to guide its application. The codebook was validated and refined using four additional transcripts, selected randomly, to ensure its applicability. We used sentences as the coding unit, segmenting the data into manageable and interpretable units for analysis [20]. While this approach provided structure, the meaning of some sentences depended on their surrounding context. To address this, coders were trained to consider the preceding and following sentences when interpreting ambiguous or incomplete ideas, ensuring accurate and context-sensitive coding. Two coders were trained using a separate subset of transcripts, which was randomly selected from the dataset. Inter-rater reliability (IRR) was calculated using Krippendorff’s Alpha, yielding substantial agreement ($\alpha = 0.88$) at the third round (11.76% of the data for each round). Discrepancies during coding were collaboratively reviewed and resolved through consensus discussions facilitated by the first author. Following this, the final codebook was applied independently by the two coders to the remaining data. In addition, researchers recorded observational notes during the content analysis process. These notes captured qualitative detail about group dynamics, and provided illustrative examples to contextualize and enrich the quantitative findings.

For each interaction form (code), we first calculated the frequency of each code within each group. Next, we computed the proportion of each code’s frequency relative to the total frequency of all interaction forms within the group, providing a normalized way to investigate group dynamics. This normalization ensures that differences in group size or total interaction count do not bias the results, allowing for a more accurate comparison of group dynamics. To analyze these proportions, we employed a GLM with a binomial distribution and logit link function. In the model, group type and familiarity were included as fixed effects to examine their influence on the distribution of interaction forms.

Besides that, the analysis of alternative rankings and the reasoning derived from the group transcripts is presented separately in another work.

Table 1 Summary of the collected data, measures, and analysis models.

Research Questions	Measure	Data	Analysis Model
Perceived Experiences (RQ1)	Comfort Inspiration Satisfaction	Questionnaire response (Q1) Questionnaire response (Q2) Questionnaire response (Q3)	Mixed-Effects Ordinal Logistic Model
Level of Participation (RQ2)	Level of Participation	Individual word counts	Linear Mixed-Effects Model
In-Group Range of Perceived Experiences and Participation (RQ3)	In-Group Range of Comfort In-Group Range of Inspiration In-Group Range of Satisfaction In-Group Range of Participation Proportion	Questionnaire responses (Q1) Questionnaire responses (Q2) Questionnaire responses (Q3) Individual word count as a proportion of the total group word count	Generalized Linear Model
Forms of Interactions (RQ4)	Forms of Interactions	Close code proportion of the total group codes from content analysis	Generalized Linear Model

4 Results

4.1 Perceived Experiences (RQ1)

We analyze the questionnaire data in two ways. We first analyze the data for all participants ($N = 94$), including both parents and children. We then specifically compared children in family-based groups with those in child-based groups to examine how different group compositions influenced their perceived SIP experiences ($N = 65$).

Comfort. Participants in groups with mixed familiarity reported significantly lower comfort in sharing ideas (Q1) compared to groups where all members knew each other ($\beta = -1.599$, $SE = 0.775$, $z = -2.062$, $p = 0.039$), while there were no statistically significant differences in comfort level between family-based and child-based groups. Specifically focusing on children's experiences, neither group type nor familiarity had a significant effect on comfort levels.

Inspiration. In terms of how group composition influenced considering issues in a new way (Q2), neither group types nor familiarity significantly influenced participants' responses. Looking at children's experiences, children in child-based groups were more likely to report considering new ideas ($\beta = 1.285$, $SE = 0.553$, $z = 2.325$, $p = 0.020$) compared with those in family-based groups, while familiarity did not show a significant effect.

Satisfaction. For levels of satisfaction with the final results (Q3), child-based groups reported significantly higher satisfaction scores than family-based groups ($\beta = 2.270$, $SE = 0.744$, $z = 3.050$, $p = 0.002$), while the impact of familiarity remained non-significant. In terms of children's experiences, similarly, participants in child-based groups reported significantly higher satisfaction scores compared to those in family-based groups ($\beta = 2.743$, $SE = 0.733$, $z = 3.743$, $p < 0.001$). Familiarity did not have a significant effect on satisfaction.

4.2 Level of Participation (RQ2)

Level of Participation. To examine participation level, we calculated the word count for each participant as a measure of their verbal contribution. Our analysis showed that for all participants ($N = 94$), neither group type nor familiarity had a significant effect on overall participation levels. Focusing specifically on children's

participation ($N = 65$), we found that children in child-based groups spoke significantly more than those in family-based groups ($b = 256.27$, $SE = 111.24$, $z = 2.30$, $p = 0.021$). Familiarity, however, was not significantly associated with children's word count.

4.3 In-Group Range of Perceived Experiences and Participation (RQ3)

In-Group Range of Perceived Experiences. To understand whether some group members had a significantly worse experience than others, we examined the range of questionnaire responses within each group. In terms of comfort level (Q1), child-based groups showed a significantly smaller range compared to family-based groups ($\beta = -0.606$, $SE = 0.292$, $z = -2.08$, $p = 0.038$), while familiarity had no significant effect. For inspiration (Q2), neither group type nor familiarity significantly affected the range. For satisfaction (Q3), child-based groups again showed a smaller range compared with family-based group ($\beta = -0.717$, $SE = 0.286$, $z = -2.505$, $p = 0.012$), and familiarity remained non-significant. These results suggest that child-based groups had more consistent experiences in comfort sharing and overall satisfaction, with no clear influence of group familiarity on considering issues in a new way.

In-Group Range of Participation. For in-group range of participation (shown as personal word count proportion of the total group word count), the results showed a significant effect of group type, with child-based groups exhibiting a smaller range of the proportion of word count compared to family-based groups ($\beta = -0.351$, $SE = 0.062$, $z = -5.674$, $p < 0.001$). Mixed-familiarity groups showed a smaller range compared to fully-familiar groups ($\beta = -0.283$, $SE = 0.099$, $z = -2.859$, $p = 0.004$). These findings suggest that child-based groups and mixed-familiar groups are associated with smaller variability in participation.

4.4 Forms of Interaction (RQ4)

We identified various interaction types during the SIP activities, including task-related, social-emotional, facilitator, and off-task behaviors. The categories, along with their definitions and examples, are presented in Table 2.

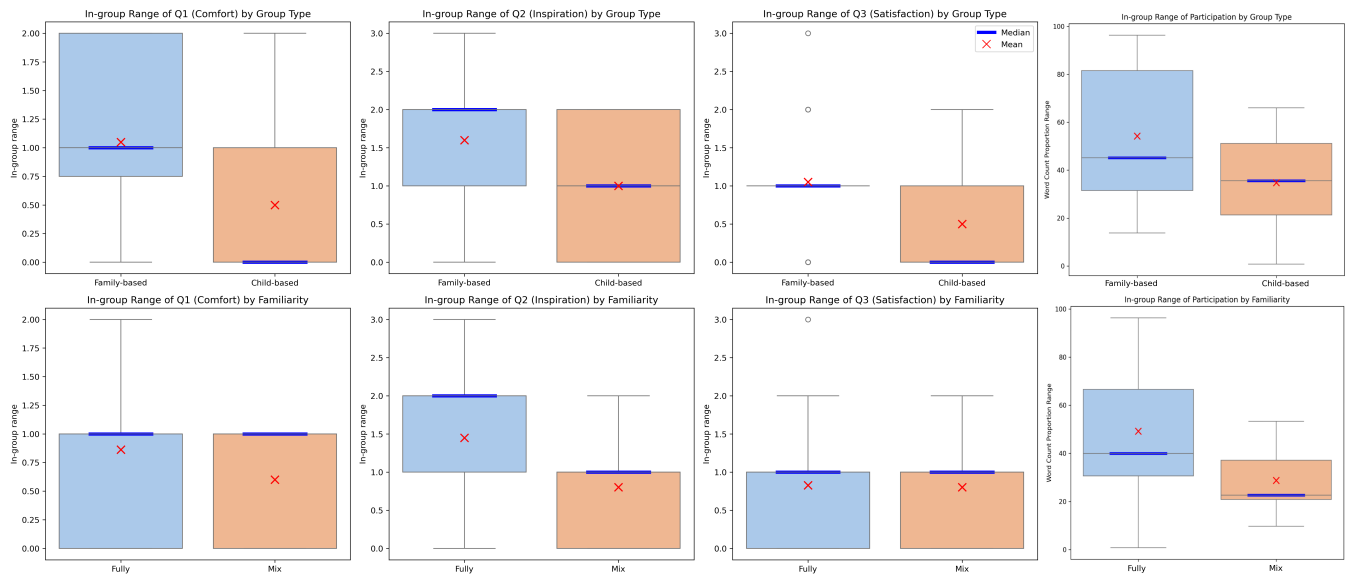


Figure 2: Distribution of in-group range for perceived experiences and participation based on group type (Family-based vs. Child-based) and familiarity (Fully vs. Mix). Each boxplot shows the median, IQR, and variability within the groups. The plots allow us to compare how group type and familiarity influence the variability, with narrower ranges suggesting more consistent responses within a group, while wider ranges imply greater variability in group responses.

Verbalizing Reasoning. Participants in child-based groups demonstrated a significantly higher proportion of verbal reasoning interactions during group discussions compared to family-based groups ($\beta = 0.784$, $SE = 0.079$, $z = 9.982$, $p < 0.001$). Participants in mixed-familiarity groups exhibited a significantly lower proportion of verbal reasoning interactions ($\beta = -0.686$, $SE = 0.114$, $z = -6.001$, $p < 0.001$). For example, in child-based group R12, two children who knew each other consistently explained their reasoning after placing each block, providing detailed justifications for their actions. When one sought opinions, the other always responded with well-thought-out reasoning. For instance, when discussing the priority of "knowledge for brain development," ID14 asked, "Is that always good information, thought?" ID13 replied, "Either it's a lot of lies," and used a mobile game as an analogy to support their argument about games in VR: "When I was playing Cells 13, I didn't experience any brain development from it because I didn't learn anything."

Minimal Acknowledgments. The proportion of minimal acknowledgments was significantly lower in child-based groups ($\beta = -0.613$, $SE = 0.180$, $z = -3.414$, $p = 0.001$) and higher in mixed-familiarity groups ($\beta = 0.904$, $SE = 0.208$, $z = 4.340$, $p < 0.001$). We found that participants, especially the children in the family-based groups or in mixed-familiarity groups, commonly used brief responses such as "Yeah," "I agree," or "OK" to respond their group members' questions or suggestions without providing further elaboration. For example, in family-based group F5, the parent (ID37) repeatedly solicited opinions, asking, "For VR, what do we think is important? ...Do you think maybe effects on relationships is an important one?" However, the child (ID38) responded simply with, "Yeah, probably." When ID37 followed up with another question,

"Yeah, what about these other ones?" ID38 still did not engage in the discussion and replied with a brief, "No."

Soliciting Opinions. Child-based groups had a significantly lower proportion of asking open-ended questions to get others' opinions ($\beta = -1.291$, $SE = 0.209$, $z = -6.192$, $p < 0.001$), while familiarity was not significant ($p = 0.784$). We found that in most family-based groups, children had a relatively low sense of engagement, so the majority of soliciting opinion questions were asked by parents. For example: "Do you feel like we need to add something else?" or "What do you think?" (ID45). In contrast, in child-based groups, while soliciting opinions did occur, children were more inclined to directly express their own thoughts rather than ask questions to engage others.

Building Consensus. Participants in the child-based groups had a significantly lower proportion of using questions to establish agreements compared with the family-based groups ($\beta = -0.592$, $SE = 0.182$, $z = -3.258$, $p = 0.001$), while familiarity showed no significant influence ($p = 0.721$). Similar to soliciting opinions, parents also used closed questions to encourage their children to participate in discussions. For example, "That could be, that would be technology, yeah, and another family, yeah?" (ID18) or "Probably effects on learning is important to learn more about, right?" (ID37). Similarly, in child-based groups, while guiding consensus did occur, children were more inclined to directly express their own thoughts rather than asking questions.

Task Management. Participants in the child-based groups exhibited significantly lower levels of task management compared to family-based groups ($\beta = -0.565$, $SE = 0.181$, $z = -3.123$, $p = 0.002$). Interestingly, mixed-familiarity groups demonstrated significantly lower task management compared with groups where

Table 2 Categories, codes, definitions, and examples of interaction

Category	Code	Definition	Examples
Task-Related Interactions	Verbalizing Reasoning	Sharing reasoning or justification for preferences.	Happiness, otherwise we don't really do this. (ID3) I put effects on behavior, happiness, society, and physical health on the top because these are all things that I think could be negatively affected by VR. (ID47)
	Task Focus Without Explanation	Contributions to task content without reasoning or elaboration.	Society. (ID2) I might say here. (ID8) I feel like relationship would be high. (ID76)
	Minimal Acknowledgments	Including simple verbal acknowledgments or affirmations to signal agreement/disagreement without further elaboration.	No, not as much. (ID37) Yeah. (ID73) I think so. (ID43)
	Soliciting Opinions	Asking open-ended questions to gather input or perspectives without aiming to resolve differences.	What do you want under this? (ID83) There's anything else that we should learn about? (ID1) Why are you putting more over here? (ID7)
	Building Consensus	Using closed-ended questions to align views and establish agreement on the current decisions regarding the board.	So you think the most needed knowledge about VR would be physical health first? (ID1) This should be family? (ID76) Do you guys agree with what I did over here? (ID82)
	Task Management	Managing time, goals, or group processes to stay on task.	For each one of these, we need to put a corresponding. (ID8) Is there anything that's missing on this card? If so, you're welcome to write it on a blank tile and then kind of put it somewhere else just like that. (ID83)
Social-Emotional Interaction	Encouragement	Encouraging remarks that support task completion or others' contribution.	Exactly, you're fine! (ID92) It's here, good job. (ID37) Okay, yeah, this is good, pretty nice. (ID87)
	Humor Complaints	Laughing, joking. Expressions of dissatisfaction.	Bang, you want to put bang, bang, other family experiences (laugh)? (ID41) This is boring... I'm not even doing anything. (ID6) I'm literally just sitting there playing with these blocks. (ID6) I thought we were going to be doing VR, not talking about it. (ID6)
Facilitator Interaction	Facilitator Involvement	Facilitator actions to clarify tasks, answer questions, or engage participants.	Do you want to weigh in? (Researcher) Those are optional. I will say you don't need to fill them out, especially if you feel like the magnets on the board capture all of your concerns. (Researcher)
	Off-Task Interaction	Non-task-related group dynamics, like small talk or casual interactions.	Oh, sorry (about dropping the block). (ID93) That fan feels so good. (ID2)

members knew each other ($\beta = -0.710$, $SE = 0.343$, $z = -2.072$, $p = 0.038$). We observed that parents in the family-based groups took on most of the task management responsibilities. For example, they would provide guidance on how to proceed with the task, "Feel free to rearrange and we'll go from there" (ID19), or reiterate activity rules to the children, such as "So it's kind of you to rate it from highest to lowest" (ID1). Parents were also the ones who sought clarification from the researchers to ensure understanding of the rules, as seen in questions like, "So these (tiles) are going to just one side or both sides?" In contrast, in mixed-familiarity groups, we observed that task management occurred less often, and facilitators took that role. Children in these groups rarely took the initiative to manage the progress of the tasks. For instance, in R2, when the group fell silent, the researcher intervened by saying, "Ok, let's move onto comfort for long-term use." Later, another pause

occurred, and the researcher prompted again, "Okay, we're gonna run through this, so physical movement."

Encouragement. Child-based groups had a significantly lower proportion of encouragement than family-based groups ($\beta = -0.980$, $SE = 0.273$, $z = -3.590$, $p < 0.001$), while familiarity showed a marginal trend but no significant impact ($p = 0.073$). Similar to the reasons that family-based groups had more interactions in soliciting opinions and building consensus, parents used more encouragement to motivate their children to contribute. For instance, even though ID86 (C) rarely spoke during the discussion and only participated by moving the tiles, ID87 (P) still said, "Okay, yeah, this is good, pretty nice."

Humor. Child-based groups had a significantly higher proportion of humor and a more relaxing atmosphere ($\beta = 2.478$, $SE = 0.871$, $z = 2.844$, $p = 0.004$), but familiarity was not significant ($p = 0.216$).

For example, in child-based group R34 (mixed-familiarity), ID93 stated, "A six-year-old shouldn't be playing a game with a 30-year-old." ID92 responded, "Download the game, bro. You're playing. Can you play, let's say, did you play GTA? But like, before you were 16?" To which ID93 jokingly said, "Before I was 10." ID92 said, "Exactly (laugh)."

Facilitator Involvement. Mixed-familiarity groups engaged facilitators more frequently than groups in which all members knew each other ($\beta = 0.771$, $SE = 0.138$, $z = 5.591$, $p < 0.001$). Group type showed no significant effect ($p = 0.807$). For example, in one mixed-familiarity group (child-based group R23), a facilitator intervened to refocus the group on the task because the discussion had stalled entirely. The researcher attempted to solicit opinions by asking, "Okay, so you can ask your partner today: Do you agree on this order? Do you want to make some rearrangements? No?" Only one child responded, and their reply was directed solely to the researcher. To encourage engagement, the researcher prompted further by saying, "You can talk to the other person to let them know your reasoning." Notably, in family-based groups, we also observed some facilitator involvement. In addition to providing encouragement and soliciting opinions for quieter groups, there were numerous instances of task management behaviors. For example, some parents frequently asked the researcher to clarify the ranking rules (e.g., whether it was permissible to assign equal rankings to two items).

Neither group type nor familiarity had a significant impact on the proportions of the rest of the codes (**Task Focus Without Explanation, Complaints and Irrelevant Interaction**).

5 Discussion

5.1 Principal Results

As shown in Table 3, our results show several patterns associated with child-based groups. Children in these groups reported higher satisfaction with the outcomes, participated more actively, and perceived their groups as more effective in generating new ideas than children in family-based groups. Child-based groups also showed more consistent experiences, as reflected in smaller in-group ranges for comfort and satisfaction. Fully familiar groups demonstrated benefits in comfort, verbal reasoning, and reduced facilitator involvement. However, familiarity showed a mixed pattern for participation balance: mixed-familiarity groups had a smaller in-group range of participation, even though they also showed lower comfort and less verbal reasoning.

Our analysis of interaction forms (Table 4) revealed distinct patterns. Family-based groups relied on parents for facilitation, including soliciting opinions, building consensus, and managing tasks. However, children in these groups were less engaged, with fewer instances of verbalized reasoning and more minimal acknowledgments. In contrast, child-based groups engaged in deeper discussions and maintained a lighter, more enjoyable atmosphere with frequent humor. Mixed-familiarity groups struggled with effective discussions, exhibiting more minimal acknowledgments, less task management and less verbalized reasoning. Facilitators also had to intervene more frequently to maintain engagement and direction.

5.2 Implications for Research

Here, we provide guidance for employing SIPs in future research.

5.2.1 Balancing Group Composition and Practical Constraints. The findings suggest that using SIP in child-based groups with fully familiar participants offers favorable outcomes for participants' discussion experiences. However, logistical constraints often make it challenging to avoid mixed-familiarity groups, especially in large-scale public settings or diverse community contexts. In such cases (including our case), mixed-familiarity groups can experience lower comfort levels and reduced in-depth interactions, with participants relying more on minimal acknowledgments and facilitator guidance. To mitigate these challenges, we recommend the following strategies: (1) Facilitators should actively bridge gaps between unfamiliar participants by encouraging interaction, clarifying and managing tasks. (2) Including brief team-building exercises at the beginning can help establish rapport and improve group cohesion. (3) Prompts should encourage verbal reasoning and equal contributions to counteract the tendencies toward minimal acknowledgments and disengagement in mixed-familiarity groups.

5.2.2 Use SIPs to Supplement In-Depth Engagements with Children. SIPs are a short-form technique to help researchers cast a wider net on children or family opinions. It is not meant as a replacement for in-depth methods like semi-structured interviews [1], contextual inquiry [44], or participatory design [30]. Instead, SIPs can complement these methods by helping researchers test, compare, and prioritize candidate directions that emerge from a smaller number of deeper engagements. The first step of a SIP is identifying alternatives, which requires sufficient prior work to serve as a foundation for this process. We encourage researchers to use SIPs to complement rather than replace other stakeholder engagement strategies.

5.2.3 Consider Group Compositions in Context. Our results showed that the child-based, fully familiar grouping offers advantages in many aspects. This aligns with prior research in group dynamics [16, 25, 46], which has shown that grouping participants homogeneously by shared roles or expertise fosters a sense of cohesion and allows for more open discussions. For instance, in contexts where stakeholders hold distinct but complementary roles (e.g., designers vs. engineers, parents vs. children), grouping by role may help to clarify priorities and ensure that each stakeholder's voice is equally valued [15, 43, 45] and also have a deeper discussion. However, if roles (or types of homogeneity) are less clearly defined to achieve the research goal, this strategy may not yield the same benefits. Previous studies in participatory methods [7, 40] have highlighted the importance of context in shaping group dynamics and group outcomes. Different contexts—such as professional settings, or educational institutions—may require different group compositions to optimize outcomes. For example, in working context, mixing roles or levels of familiarity could stimulate more creative or innovative solutions [47] (which is not validated in our study). Future work should explore how SIP's effectiveness varies across diverse contexts and group compositions, identifying the conditions under which different composition strategies are most advantageous.

Table 3 Summary of results on comfort, inspiration, satisfaction, participation levels and in-group range for Group Type [C] (Child-based) and Familiarity [M] (Mixed-familiarity). "-" means no statistically significant effect on this measure.

Measure	Description	Group Type [C]	Familiarity [M]
All Participants (N = 94)			
Comfort	All participants questionnaire response (Q1)	–	Negative
Inspiration	All participants questionnaire response (Q2)	–	–
Satisfaction	All participants questionnaire response (Q3)	Positive	–
Participation Levels	All participants individual word counts	–	–
Children Participants (N = 65)			
Comfort	Children participants questionnaire response (Q1)	–	–
Inspiration	Children participants questionnaire response (Q2)	Positive	–
Satisfaction	Children participants questionnaire response (Q3)	Positive	–
Participation Levels	Children participants individual word counts	Positive	–
All Groups (N = 34)			
In-Group Range of Comfort	In-group range of questionnaire responses (Q1)	Negative	–
In-Group Range of Inspiration	In-group range of questionnaire responses (Q2)	–	–
In-Group Range of Satisfaction	In-group range of questionnaire responses (Q3)	Negative	–
In-Group Range of Participation	In-group range of individual word count proportions	Negative	Negative

Table 4 Summary of results on forms of interaction for Group Type [C] (Child-based) and Familiarity [M] (Mixed-familiarity). "-" means no statistically significant effect on this code.

Category	Code	Group Type [C]	Familiarity [M]
Task-Related Interactions	Verbalizing Reasoning	Positive	Negative
	Task Focus Without Explanation	–	–
	Minimal Acknowledgments	Negative	Positive
	Soliciting Opinions	Negative	–
	Building Consensus	Negative	–
Social-Emotional Interaction	Task Management	Negative	Negative
	Encouragement	Negative	–
	Humor	Positive	–
	Complaints	–	–
Facilitator Interaction	Facilitator Involvement	–	Positive
Off-Task Interaction	Irrelevant Interaction	–	–

5.2.4 Explore Recruitment Strategies for SIP's Scalability. In line with research on recruitment in participatory methods, which highlights how the mode of recruitment can significantly influence participation rates, demographics, and data quality [8, 26, 38], future studies should investigate alternative recruitment strategies, such as online and hybrid approaches. Online recruitment, for example, has been shown to lower barriers to participation, especially for stakeholders who may be geographically dispersed or time-constrained [8]. However, it also introduces challenges related to sustained engagement and the depth of interaction [2]. Given that prior work suggests a potential trade-off between ease of access and the quality of insights in virtual settings [2, 8], future research should explore how recruitment strategies might be impacted or optimized to maintain the quality of outcomes and experiences of SIP, which is good for scaling SIP to different stakeholder populations.

5.3 Limitations and Future Work

This study has several limitations that articulate possible directions for future work. First, although we position SIPs as potentially relevant for a variety of stakeholders (especially children) and contexts, this study only piloted them with families reflecting on VR. We encourage researchers who conduct SIPs in other contexts to share their results and reflections on the applicability of this method to their context and any specific adaptations they need to make to the approach.

Second, our original research plan focused specifically on articulating differences between family-based and child-based groups. Further stratification of child-based groups into levels of familiarity emerged as a post-hoc artifact of our recruitment strategy. As such, our study was underpowered on this nested variable, and we are only able to provide initial insights into the effects of member familiarity on SIP experiences. This is clearly a potential area for

future study, and we hope our initial results could be used to guide the power analysis to properly power such a study.

In addition, family-based groups should not be always interpreted as a weaker design choice. Rather, they support different research goals. In our data, family-based groups showed more task management, opinion solicitation, encouragement, and consensus-building, often led by parents. While these dynamics may reduce children's independent verbal contribution, they can also reveal how families negotiate technology priorities together. Family-based groups may therefore be especially useful when the research goal is to study parent-child negotiation, family norms, or intergenerational concerns around technology use.

Finally, we encourage future work to explore more granular group dynamics. For instance, homogeneity can be defined along multiple dimensions. In our study, child-based groups were composed entirely of children, yet prior work has noted that other factors such as age, gender, or ethnicity [16, 32, 37] could further differentiate the degree of homogeneity or heterogeneity within these groups, thus affect the process and outcomes. Similarly, in heterogeneous groups, like the family-based groups in our study, examining the influence of pre-existing relationships [5] on SIP experiences would provide valuable insights. Additionally, more factors like task complexity, group size, level of familiarity, and expertise levels [31] could be investigated to understand how they affect the prioritization process and provide more nuanced guidelines for optimizing group structures in diverse participatory research contexts.

6 Conclusion

This study used Stakeholder-Informed Prioritization (SIP) as a practical method for rapidly gathering feedback from multiple child-involved small groups to examine group composition effects. Our findings suggest that child-based grouping and participant familiarity shaped SIP experiences in different ways. Child-based groups showed more favorable patterns in satisfaction, child participation, participation balance, and verbal reasoning, while fully familiar groups showed higher comfort, more verbal reasoning, and less facilitator involvement. Future work should examine how SIP can be adapted across settings and recruitment contexts, and test additional composition variables such as homogeneity vs. heterogeneity, familiarity gradients, and alternative recruitment strategies.

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A SIP Materials Used in the Study

A.1 First SIP Topic: Most Needed Knowledge about VR

Researcher script for introducing the first SIP activity: “First, we would like you to consider what is most important to know about Virtual Reality to decide whether it’s right for your family to use it at home. For example, you might want to know what data is collected about you by these systems, or you might want to see evidence on how long-term VR use affects children’s brain development, or you might want to hear how things went for other families who have used VR. We’d like you to discuss and position these magnets in this board area with the most important things to know towards the top—you can have ties by putting things on the same level. If there’s something we didn’t think of, you can write a new idea on these magnets.”

Alternatives provided the first SIP activity:

- Effects on Relationships
- Effects on Long-term Happiness
- Effects on Learning
- Effects on Brain Development
- Effects on Physical Health
- Effects on Behavior
- What Data is Collected
- Other Families’ Experiences
- Effects on Society

A.2 Second SIP Topic: Most Needed Advances, Tools, & Content for VR

Researcher script for introducing the second SIP activity: “We would like you to consider what features or advances may be most important for technologists to build next for you to be willing to use VR at home. For example, you might think that what matters most is that VR is comfortable for anybody to use for a long time without dizziness or discomfort, or you might think that what’s most important is good educational content or easy-to-use parental controls. Again, we’d like you to discuss and position these magnets in this board area with the most important things to know towards the top.”

Alternatives provided in the second SIP activity:

- Fun Social Experiences
- Educational Content
- Comfort for Long-term Use
- More Realistic Graphics
- Physical Movement
- Safety Monitoring (e.g., no bullying)
- Privacy Protections
- Price & Availability
- Parental Control Tools

B VR Scenarios Used in the Study

We used the following two VR scenarios displayed as videos, which were borrowed from a prior work on VR ethics for children [23].

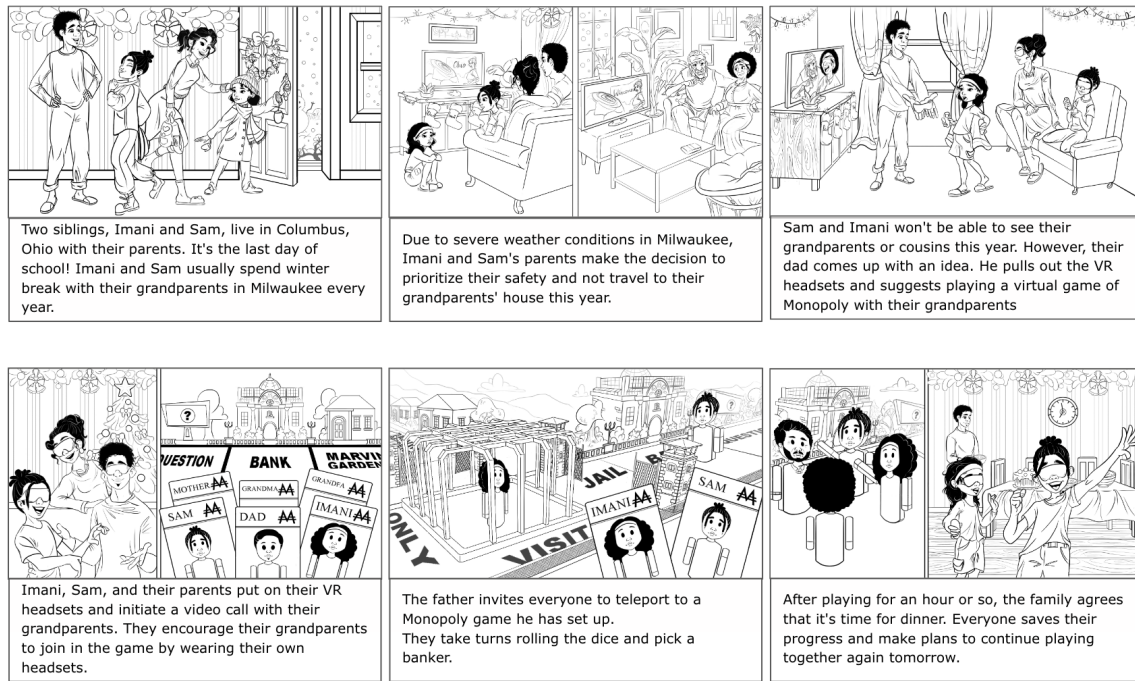


Figure 3: Family engaging together in VR gameplay.

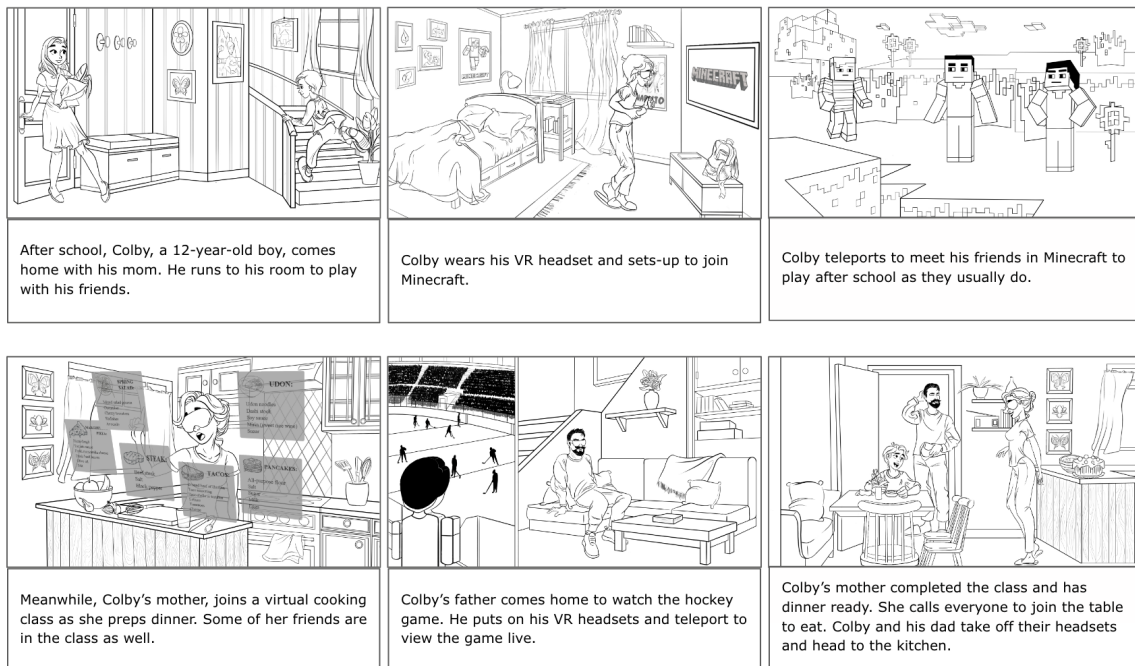


Figure 4: Family members individually immersed in separate VR experiences while together physically.